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Toxicity of seed borne storage fungi of Cowpea(*Vigna sinensis* L) II- Quantitative change in photosynthetic pigments and sugar in the seedlings.

Monalisha Saha[a]* and H.B.Sahu[a]
Microbiology and Plant Pathology Laboratory, Ranchi University, Ranchi, Jharkhand, India.

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Abstract : Total chlorophyll, carotene , xanthophull and total sugar were less in the seedlings of Cowpea raised from the seeds stored with *Aspergillus flavus, A. niger* and *Fusarium moniliforme*. As the RH of storage was raised from 60 to 80%, there appeared proportionate decrease in the noted bio-chemicals. *A. flavus* proved as the most deleterious followed in succession by *A. niger* and *F. moniliforme*. Glucose , fructose and sucrose that were detected as individual sugars in the seedlings were found distinctly less due to the seed borne fungi when stored with seeds at 80% RH.

Key Words: Cowpea seed, storage fungi, photosynthetic pigments, sugars in seedlings, deleterious, carotene , xanthophylls.

INTRODUCTION

The seed borne storage fungi have been held responsible in deteriorating the seeds with the consequence of their reduced germination (Christensen, 1972[1]; Khan and Siddiqui, 1980[2]; Prasad, 1982[3]; Sao et al, 1989t ; Kiran 2009u). In recent years, besides chasing the biochemical alterations in the seeds due to storage fungi, these have also been explored in the seedlings (Kishor and Prasad, 1989v ; Prasad et al 2011w).Like the loss in the photosynthetic pigments in the host due to the pathogens, the storage fungi have also been known to decrease the amount of photosynthetic pigments and sugars in the seedlings of a few crop plants (Kishor and Prasad 1989v ; Sao et al 1989t ; Diwakar and Prasad1997x ; Rajnish et al 2003y).

In the present paper attempts have been made to see the effect of seed borne storage fungi of Cowpea (Vigna sinensis L.) seeds on the quantitative change in photosynthetic pigments and sugars in the seedlings.

*Corresponding author :
Phone: 09162274554, 09939469268
E-mail : saha.monalisha85@gmail.com

MATERIALS AND METHODS –

The seeds of Cowpea (*Vigna sinensis* L.) were collected from different blocks of Ranchi district, and fungi were isolated adopting blotter technique (Tempe1963[1]R″). Based on their higher frequency, *Aspergillus niger* Van Teigham, *A. flavus* Link ex Fries and *Fusarium moniliforme* Sheldon were selected from the total isolated lot of fungi. These were culture on Czepek Dox Agar slants at 28± 1ÚC for 7days. 20 gm seed lot of Cowpea possessing 100% germinability and containing 7.82% moisture were surface sterilized with 0.1% mercuric chloride solution for 1min. This lot was infested with 1ml of spore suspension containing 10 t spore/ml of the noted fungi prepared in 5% of Tween 20. The number of spores was counted with the help of haemocytometer. The infested lots were stored over glycerol solution to maintain 60, 70 and 80% RH. at 30 ± 1R″C.in sealed desiccators (Braun and Braun1958[11])for 30 days.

CULTURE OF THE SEEDLINGS –

After expiry of the storage period, 20 seeds were sown nearly at equal distance, 1cm.deep in autoclaved garden soil taking in earthern pots of 25X20X15cm dimension. For each treatment triplicate of pots were

prepared. The surface of the soil was covered with 1cm thick sterilized cotton wool. Light watering was maintained alternate day. After germination of the seed on the fifth day, cover of the cotton was removed and the pots were shifted to the garden and the seedlings were permitted to grow for 10 days.

ESTIMATION OF THE PHOTOSYNTHETIC PIGMENTS–

The first leaf of the seedling was cut at 2.0pm and immediately put at -10 C in pitch dark for 30 minutes. 500gm of the leaf was weighed and soon extracted with 5ml of 80% acetone for five times in the dark. The extracts after pooling together, was centrifuged at 5000 rpm for 10 minutes and the final volume was made to 50ml with 80% acetone in volumetric flask. The absorbance was measured in spectrophotometer at 645 and 663 nm and the amount of total chlorophyll was calculated adopting Harborne (1973[12]) as noted below—

Total chlorophyll (mg/gm fresh leaf) =

$$\frac{20.2A645 + 8.02A663}{a \times 1000 \times W} \times V$$

Where – a= length of light path in the cell (1cm)
V = Volume of the extract in ml
W = Fresh weight of the leaf in gm.
A = Absorbance at noted wavelength.

CAROTENE AND XANTHOPHYLL–

5 ml of the acetone extract prepared earlier was concentrated in vacuum in the dark to 0.5 ml and 20 µl of this concentrated extract was spotted on TLC plate (250 µl thick Silica gel G as absorbant) and the separation of the pigments was made using the solvent (Petroleum ether, B.P. 90-110 C :Benzene, 50:50, v/v)for 2 hr in the dark at 25 C. The carotene (Orange in colour) and Xanthophyll (Yellow in colour) spots, that were separated, were scrapped. Silica gel containing the pigments , was centrifuged at 10,000 rpm for 10 min. and the supernatant was read at 440 nm and calculation was made as follows adopting Snell and Snell(1953[13])-

mg Carotene = Absorbance X 0.028.

Xanthophyll was eluted with 10 ml of 100% ethanol and simply absorbance was recorded at 442 nm adopting Snell and Snell (1954[1]t). Photosynthetic pigments were recorded in Table – I.

ESTIMATION OF SUGAR—

The sugar was estimated as total sugar and individual sugars in the seedlings. The residue after extraction of photosynthetic pigment was extracted by warming with 3ml of 80% ethanol four times. The ethanol extracts were pooled together and centrifuged at 10,000 rpm for 10 min. The supernatant was warmed and vacuum evaporated till the odour of ethanol ceased to emit. The liquid was made to 2ml. This stock solution of the sugar extracted from 500 mg of leaf was used for the estimation of total sugar and individual sugars.

0.5 ml of the stock sugar extract was diluted to 5ml and estimated using Anthrone reagent adopting Snell et. al.(1961[1]u). 2ml of the solution was mixed with 10ml of 0.1% Anthrone reagent and the mixture was warmed at 100R″C for 20 min. On cooling to room temperature, it was read at 625 nm and total sugar was calculated with the help of calibration curve prepared by serial dilution of 1.0% glucose solution.

Individual sugars in the stock extract of the seedlings raised from the seeds stored, were identified after their separation adopting TLC method (Stahl, 1969[1]v). 20µl of the sugar solution was spotted on two plates and both were developed ascending in the solvent Benzene: Glacial Acetic Acid: Methanol (20:20:60, v/v/v.) to the height of 10cm. Out of two plates, the sugars were detected on one plate by spraying Aniline-Diphenylamine-Orthophosphoric acid reagent(ADOA reagent), and identified from standard chromatogram prepared similarly of the known sugars. Altogether three sugars i.e. glucose, fructose and sucrose were identified in the leaf extract. For the quantitative analysis of the identified sugars, the remaining TLC plate which has not been sprinkled with the detecting reagent (ADOA) was used. The area of the sugars on the TLC plate, after comparison with the first plate was scrapped with steel scalpel and extracted three times, using 2ml of warm water at 50R″C. The extracts after their cooling was centrifuged at 10.000 rpm. The individual sugars were quantified adopting Snell et. al.(1961[1]u) as noted earlier for total sugar. In this case 1.0% of glucose, fructose and sucrose were separately used for preparing calibration curve.

RESULTS AND DISCUSSION —

It appears that the storage of Cowpea seeds with seed borne fungi *A. niger, A. flavus* and *F. miniliforme* resulted in decrease of all the photosynthetic pigments i.e. total chlorophyll, carotene and xanthophylls in their seedlings(Table -1) and maximum due to *A. flavus* followed in succession by *A.niger* and *F.moniliforme*. Also, as the RH of storage of seeds in association with the noted fungi was raised, there was proportionate decrease in the photosynthetic pigment (Table -1). The control seeds produced seedlings with negligible decrease in the photosynthetic pigments as compared to the fungus stored seeds. Total sugars follow the similar trend of decrease in seedlings after raising them from the fungus stored seeds (Table – II). Among the individual sugars glucose, fructose and sucrose were detected. These too, were less in the seedlings raised from the fungus stored seeds at 80% R.H. as compared to the control (Table III). It is obvious that the storage fungi can be held responsible for fewer amounts of the photosynthetic pigments in the seedlings suggesting either their retarded synthesis or breakdown of the pigments already synthesized in normal course. The possibility of the former fate of photosynthetic pigments seems strong due to the fact that the cotyledons of Radish (Sao et. al. 1989t) and of the mustard (Singh , 1988[1]w) remained yellow as found in natural way, and also failed to expand as cotyledonary leaf due to storage of seeds with these fungi. The seeds of mustard soaked in the metabolite of the noted seed borne fungi and acetone extract of fungus stored seeds produced seedlings having deficient amount of chlorophyll (Kumar, 2011[1]x). In contrast to these, the chlorophyll pigments have been found to be degraded in the host due to the secretion of chlorophyllase after invasion of pathogenic fungi (Goodman et.al. 1967[1]y). Thus there is difference in the effect of two categories of fungi on the degradation of photosynthetic pigments. The involvement of fungus secreted toxin in hindrance, in the synthesis of chlorophyll, carotenes and xanthophylls can be highly conjectured as the fungus free metabolites help produce restricted amount of chlorophyll in mustard seedling (Kumar, 2011[1]x). Also, Mg ++ is the integral part of chlorophylls and some other cations are required for their synthesis, As reported earlier, many cations are leached out due to denaturation of the plasma membrane from the seeds by their association with storage fungi (Kishor,1989[2]R″). The cat ions are also leached out from the root of seedlings raised from the fungus deteriorated seeds (Kishor et. al.1990[21]) Synthesis of carotene and xanthophylls might also be disrupted due to some toxic principles as noted for the chlorophylls. The secretion of toxic principles by seed borne *Aspergillus ruber* has earlier been reported in pea seeds (Harman and Nash, 1972[22]).

Scant amount of total soluble sugar in the seedlings of Cowpea following the comparable trend of the photosynthetic pigments, can naturally be assumed due to their insufficiency. The individual sugars detected were also less in the seedlings raised from the seeds stored with the fungi. Besides their suppressed synthesis due to less photosynthetic pigments in the seedlings, their excited leakage from the root (Kishor et al 1990[21]) and rapid loss as substrate due to augmented respiration in similar condition (Kishor and Prasad, 1989v ; Prasad et.al., 2011w) seem the potential factor in their diminution. The storage of Cowpea seeds with storage fungi at 60% R.H. seems less detrimental in comparison to that at 80% R.H. due to the fact that the latter creates congenial environment for luxuriant growth of the storage fungi by raising the equivalent moisture of the seeds (Bottomley et. al., 1952[23]), and consequently producing more toxins.

Table 1: Photosynthetic pigments in the seedling of Cowpea (*Vigna sinensis* L) raised from the seeds stored with seed borne fungi at varying R.H. (Expressed as mg/gm fresh leaf, Xanthphyll as O.D.).

Particulars		RH IN %			
	Seed borne fungi	60	70	80	Mean
Total Chlorophyll	*A.niger*	1.680	1.513	1.280	1.491
	A.flavus	1.489	1.395	0.951	1.278
	F.moniliforme	1.736	1.602	1.376	1.571
	CONTROL	1.962	1.946	1.921	1.943
	MEAN	1.716	1.614	1.382	
Carotene	*A.niger*	0.139	0.111	0.087	0.112
	A.flavus	0.117	0.106	0.077	0.100
	F.moniliforme	0.148	0.129	0.092	0.123
	CONTROL	0.187	0.181	0. 165	0.177
	MEAN	0.147	0.131	0. 105	
Xanthophyll	*A,niger*	0.135	0.120	0.107	0.120
	A.flavus	0.121	0.113	0. 098	0.110
	F.moniliforme	0.146	0. 139	0. 118	0.134
	CONTROL	0.165	0.158	0. 146	0.156
	MEAN	0.141	0.132	0.117	

Table II: Total sugar in the seedling of Cowpea (*Vigna sinensis* L.) raised from the seeds stored with seed borne fungi at varying % of RH(Expressed as mg./gm fresh leaf).

Seed borne fungi/Control	RH IN %			
	60	70	80	Mean
A.niger	17.62	15.27	12.22	15.03
A.flavus	15.10	13.05	10.12	12.75
F.moniliforme	20.56	18.37	14.27	17.73
CONTROL	30.72	28.82	25.27	28.25
MEAN	21.05	18.87	15.64	

Table III: Individual sugars in the seedling of Cowpea (*Vigna sinensis* L.) raised from the seeds stored with seed borne fungi at 80% RH (Expressed as mg/gm fresh leaf).

Seed borne fungi	Sugars		
	Glucose	Fructose	Sucrose
A.niger	5.12	3.89	3.21
A.flavus	4.47	3.12	2.53
F.moniliforme	6.38	4.75	3.85
CONTROL	11.13	8.82	5.32

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